PROJECT REPORT

ON

MEDICAL INFORMATION SYSTEM

Submitted in partial fulfilment of Bachelor of Computer Applications (BCA)

To

UTTARANCHAL UNIVERSITY, DEHRADUN

UTTARANCHAL SCHOOL OF COMPUTING SCIENCES

SESSION (2025-26)

Submitted To:	**Submitted by:**
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	Section – D

ACKNOWLEDGMENT

The completion of any work is always a rewarding moment, but it is never the result of individual effort alone. It is the outcome of constant support, guidance, and encouragement from many people, and I would like to acknowledge their contribution in making this project possible.

First and foremost, I extend my sincere gratitude to Prof. (Dr.) Sonal Sharma, Director of USCS, for providing such a motivating and positive environment that inspired me throughout my studies.

I am deeply thankful to Prof. (Dr.) Sameer Dev Sharma, Head of Department, USCS, for his encouragement and valuable suggestions. I also owe my heartfelt gratitude to my project mentor, Dr. Manoj Aswal, for his patient guidance, constructive feedback, and continuous support at every stage of this project, which greatly strengthened my confidence and learning.

Simran

Enroll. No. UU2509000534

BCA (2nd Sem)

Section-'D'

DECLARATION

I, Simran , hereby declare that the mini project titled "Medical Information System" submitted by me is a record of original work carried out by me under the guidance of my teacher. This project has not been submitted to any other university or institution for the award of any degree, diploma, or certificate.

I further declare that all the information and data included in this project are collected from genuine sources and are acknowledged properly wherever used.

Simran

Enroll. No. UU2509000534

CERTIFICATE OF ORIGINALITY

This is to certify that the project report titled “MEDICAL INFORMATION SYSTEM" submitted by Simran, Enroll. No. UU2509000534, in partial fulfilment of the requirements for the degree of BCA Data analysis, is a record of original work carried out by her under my supervision and guidance.

The project work was carried out during the academic year 1st and it has not been submitted previously to any other institution or university for the award of any degree or diploma.

Under the guidance of

Dr. Manoj aswal

Assistant Professor-

USCS

Uttaranchal University Dehradun

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Introduction

A Medical Information System (MIS) is a computer-based system designed to store, manage, and process healthcare data efficiently. It plays a crucial role in modern healthcare by helping hospitals, clinics, and medical professionals maintain accurate patient records and streamline their daily operations.

The system allows users to store important information such as patient details, medical history, diagnosis, prescriptions, and billing records in a digital format. This reduces the need for manual paperwork and minimizes errors, making the process faster, safer, and more reliable.

With the help of a Medical Information System, healthcare providers can easily access patient information whenever required, improving decision-making and patient care. It also helps in managing appointments, tracking treatments, and maintaining reports systematically.

This project aims to develop a simple and user-friendly Medical Information System that can handle basic hospital functions such as patient registration, record management, and data retrieval. The system ensures better organization, quick access to data, and improved efficiency in healthcare services.

Objectives

Provide easy access to progress reports for students and working professionals anytime and anywhere. Allow parents to monitor their child's academic performance through a secure login system. Maintain a centralized digital record of performance, attendance, and achievements. Reduce paperwork and manual handling of progress reports. Improve transparency between students, parents, teachers, and organizations. Ensure data security by using login IDs and passwords for authorized access. Enable quick updates of performance reports by institutions or organization

System Analysis

HARDWARE AND SOFTWARE USED

HARDWARE USED: -

- Processor intel Core i5 (14 generation)
- Storage 512 Gb SSD
- Mouse
- Laptop

SOFTWARE USED: -

- Windows OS
- Dev c ++
- Vs code
- Operating System Windows 11

Problem Statement

In many hospitals and clinics, patient records and medical data are still managed manually using paper-based systems. This traditional method is time-consuming, prone to errors, and difficult to maintain. Retrieving patient information becomes slow and inefficient, especially when dealing with a large number of records.

Manual systems also increase the risk of data loss, duplication, and mismanagement of important medical information. In addition, managing appointments, billing, and treatment records becomes complicated and unorganized.

There is a need for an efficient and reliable system that can store, manage, and process medical information in a structured and secure way. The system should provide quick access to patient data, reduce paperwork, and improve overall hospital management.

This project aims to solve these problems by developing a Medical Information System that automates data handling, ensures accuracy, and enhances the efficiency of healthcare services.

Definitions

Online Portal

A web-based application that allows patients and healthcare providers to access and manage medical information anytime and anywhere

User Authentication

User authentication is the process of verifying the identity of a user before granting access to the Medical Information System

Progress Report

A document that shows the current status of a project. It explains what work has been completed, what is still in progress, and what tasks are planned for the future

Login Interface

The login interface is the entry point of the Medical Information System where users authenticate themselves to access the system

Dashboard

The dashboard is the main interface of the Medical Information System that appears after a user successfully logs in

Database

The database is a crucial part of the Medical Information System that stores, organises, and manages all the data related to the system.

Requirement Specification

1. Functional Requirements

Functional requirements describe

1.1 User Registration

The system should allow new users (patients/doctors) to create an account by providing basic detail

1.2 User Login

The system should verify user credentials and allow only authorised users to access their accounts

1.3 User Profile Management

Each user (patient, doctor, or admin) has a profile that stores important information such as name, contact details, and other relevant data

1.4 Appointment

management Patients can

book appointments

Doctors can view and manage

appointments System should store

appointment details

1.5 Patient record management

The system should allow doctors to view and update patient medical records

1.6 Doctor management

Admin can add, update, or remove doctor information

1.7 Dashboard

The system should provide a dashboard that displays:

- Main interface displayed after user login
- Shows summary of important information (appointments, records)
- Provides quick access to different system features
- Displays data based on user role (patient, doctor, admin

2. Non-Functional Requirements

Non-functional requirements describe the quality and performance of the system.

2.1 Performance

The system should load quickly and respond efficiently to user actions.

2.2 Security

User data and login information should be protected using proper authentication and security measures.

2.3 Usability

The system should have a simple and user-friendly interface so that students can easily navigate and use the features.

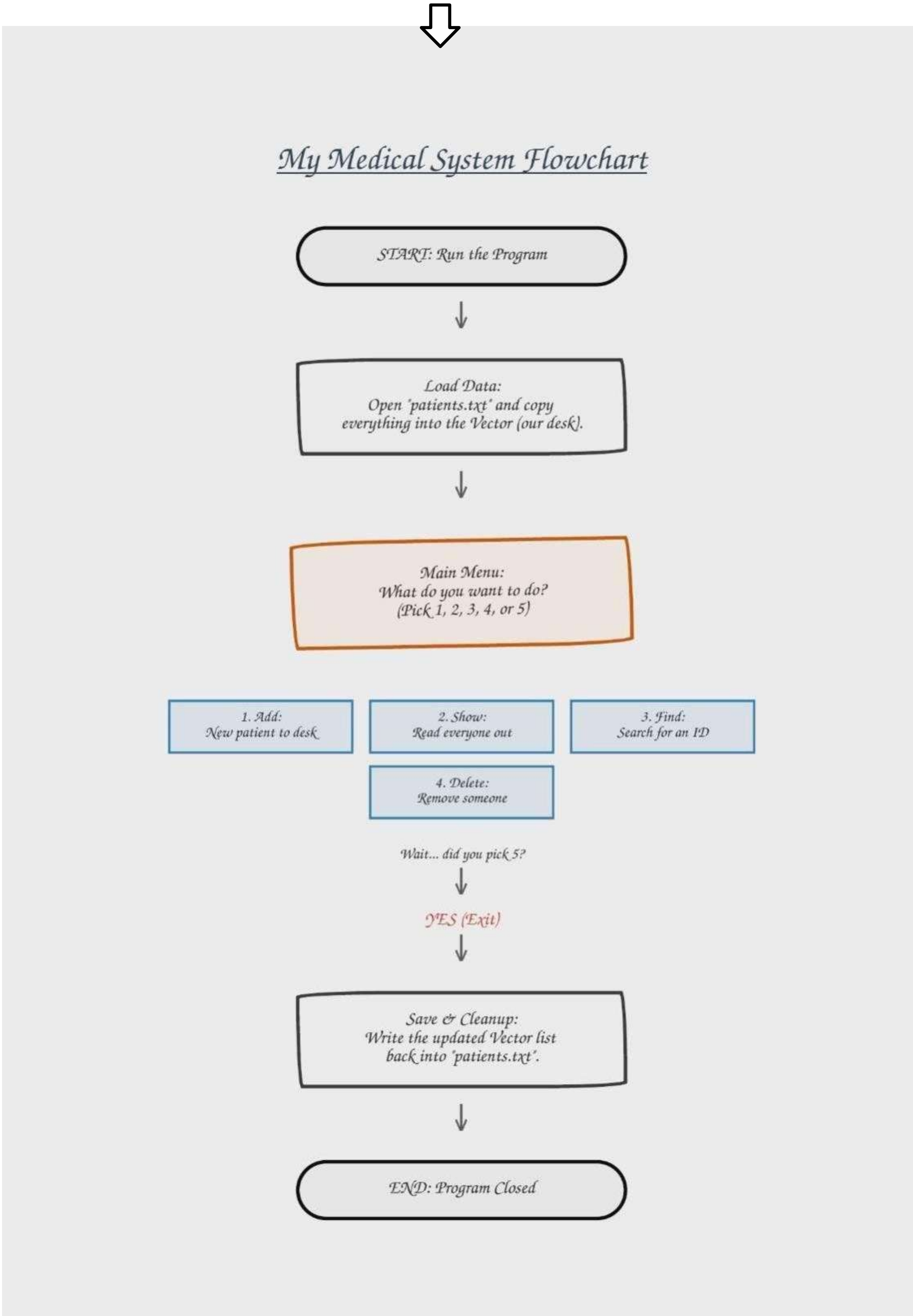
2.4 Compatibility

The system should work properly on different devices such as mobile phones, laptops, and desktop computers.

2.5 Reliability

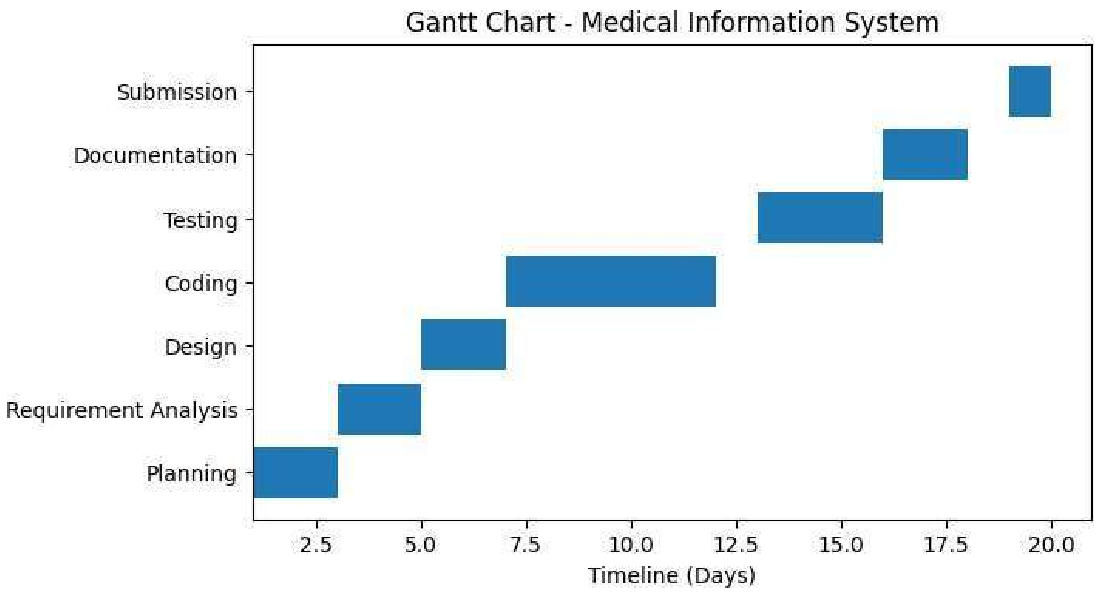
The system should store and manage user data accurately without data loss.

FLOW CHART



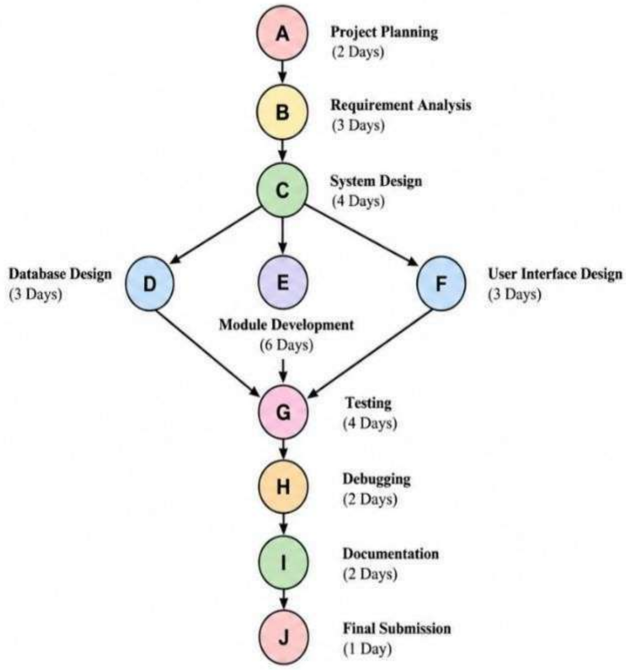
My Medical System Flowchart
START: Run the Program
Load Data:
Open 'patients.txt' and copy
everything into the Vector (our desk).
Main Menu:
What do you want to do?
(Pick 1, 2, 3, 4, or 5)
1. Add:
New patient to desk
2. Show:
Read everyone out
3. Find:
Search for an ID
4. Delete:
Remove someone
Wait... did you pick 5?
YES (Exit)
Save & Cleanup:
Write the updated Vector list
back into 'patients.txt'.
END: Program Closed

GANTT CHART

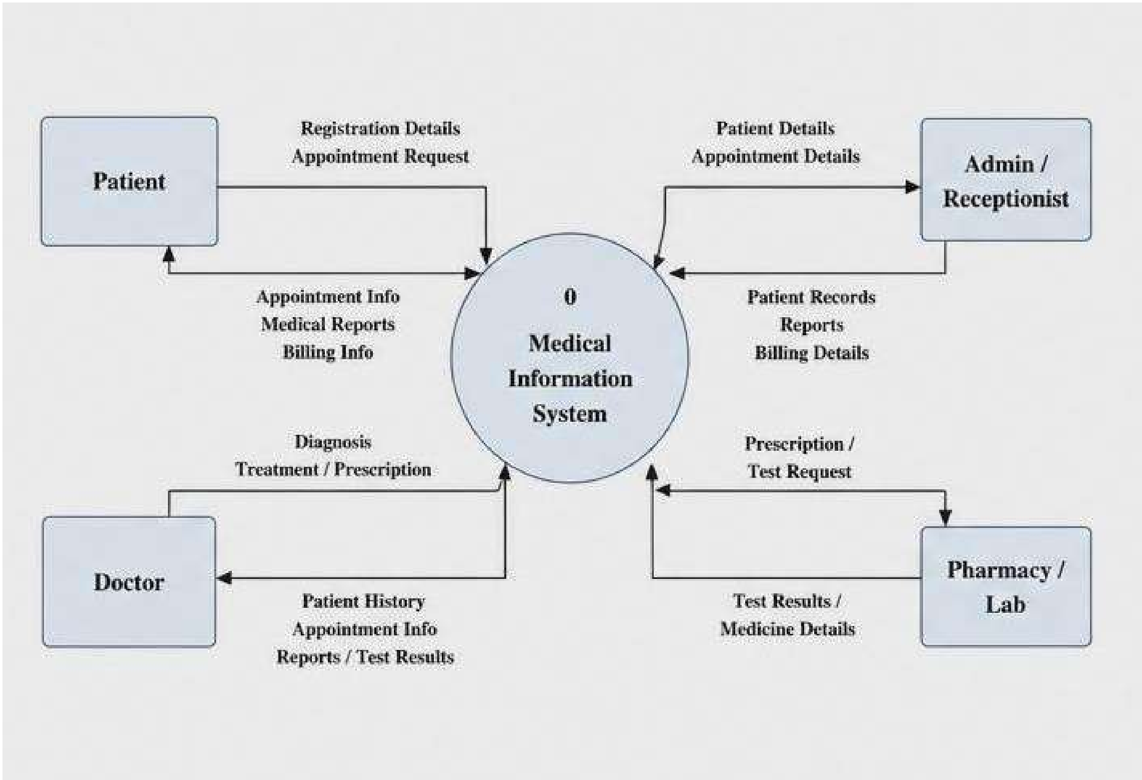

Gantt Chart - Medical Information System
Submission
Documentation
Testing
Coding
Design
Requirement Analysis
Planning
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Timeline (Days)

PERT CHART

The PERT chart below represents the sequence of activities involved in the development of the Medical Information System.

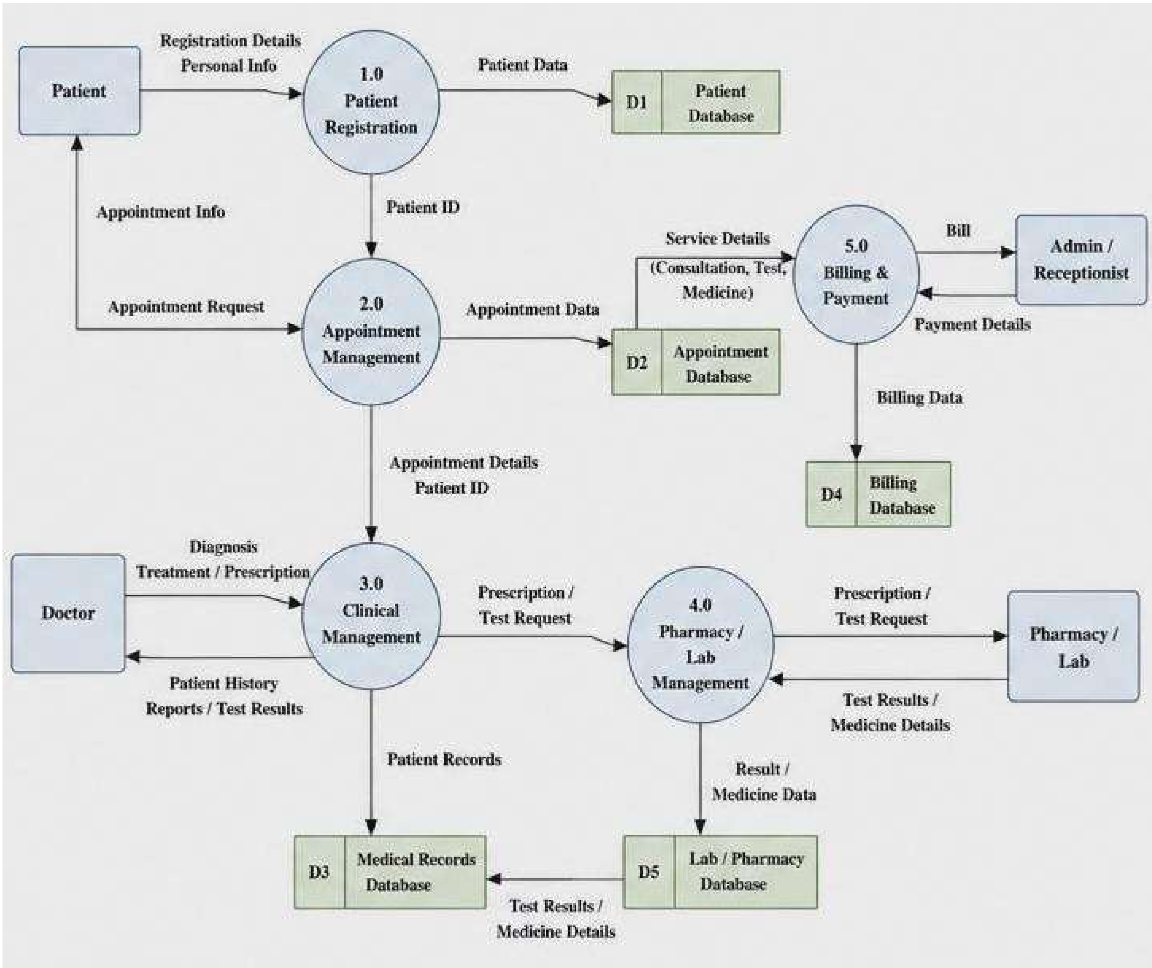


DFD Level 0

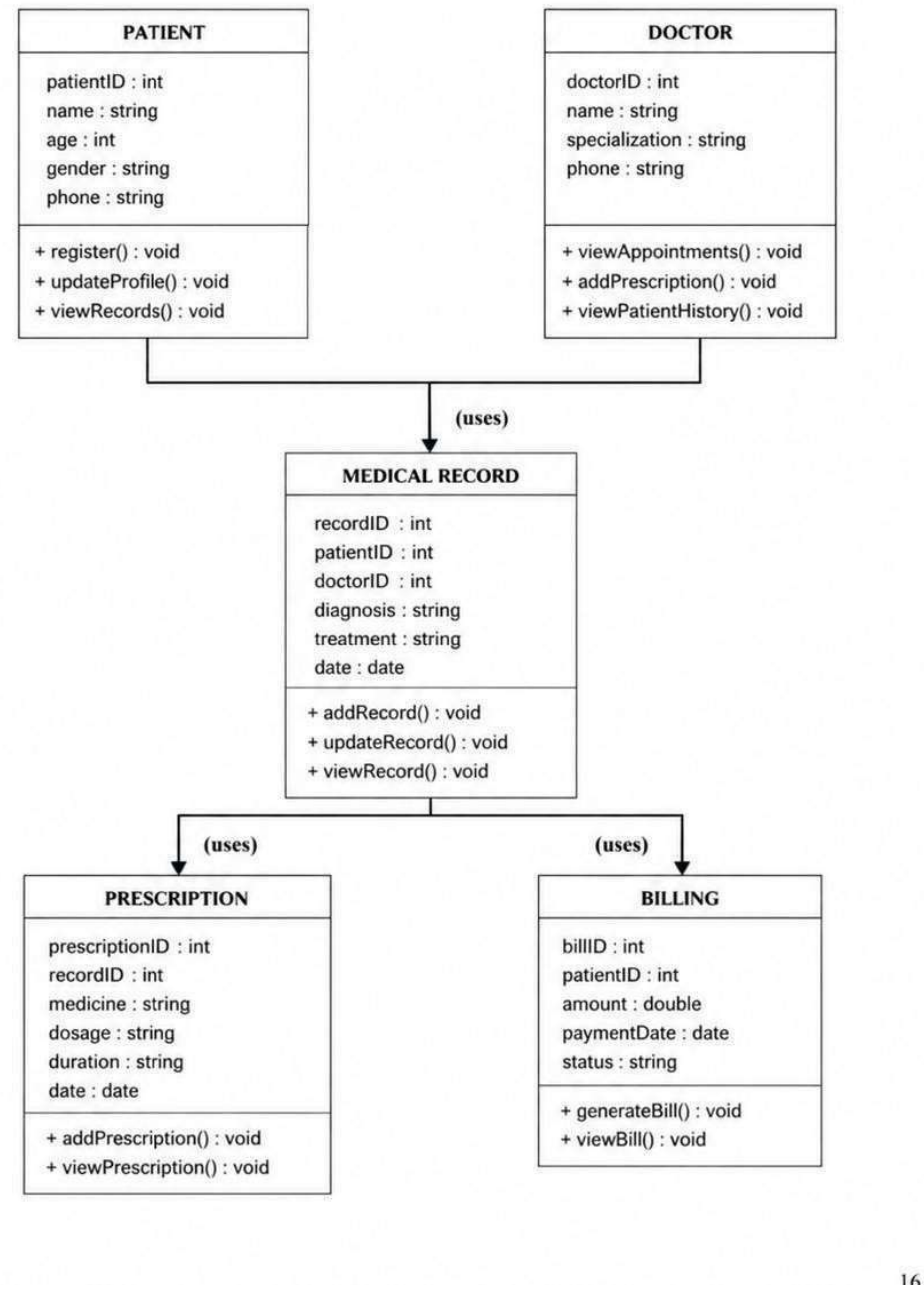


Patient
Registration Details
Appointment Request
Patient Details
Appointment Details
Admin / Receptionist
Appointment Info
Medical Reports
Billing Info
0
Medical Information System
Patient Records
Reports
Billing Details
Diagnosis
Treatment / Prescription
Prescription /
Test Request
Doctor
Patient History
Appointment Info
Reports / Test Results
Test Results /
Medicine Details
Pharmacy / Lab

DFD Level 1


Patient
Registration Details
Personal Info
1.0
Patient
Registration
Patient Data
D1
Patient
Database
Appointment Info
Patient ID
Service Details
(Consultation, Test, Medicine)
5.0
Billing &
Payment
Bill
Admin / Receptionist
Payment Details
Appointment Request
2.0
Appointment
Management
Appointment Data
D2
Appointment
Database
Billing Data
D4
Billing
Database
Appointment Details
Patient ID
Diagnosis
Treatment / Prescription
Doctor
3.0
Clinical
Management
Prescription /
Test Request
4.0
Pharmacy /
Lab
Management
Prescription /
Test Request
Pharmacy /
Lab
Patient History
Reports / Test Results
Test Results /
Medicine Details
Patient Records
Result /
Medicine Data
D3
Medical Records
Database
D5
Lab / Pharmacy
Database
Test Results /
Medicine Details

CLASS DIAGRAM



Future Scope

The Medical Information System can be further enhanced and expanded with advanced features to improve its functionality and usability.

- The system can be upgraded to a web-based or cloud-based platform so that data can be accessed anytime and from anywhere.
- Integration with mobile applications can allow doctors and patients to access records, book appointments, and receive notifications easily.
- Advanced features like online appointment booking and SMS/email alerts can be added for better communication.
- The system can include electronic health records (EHR) for maintaining complete and detailed patient history.
- Implementation of data security techniques such as encryption and user authentication can improve data safety.
- It can be integrated with laboratory and pharmacy systems to manage test reports and medicine records efficiently.
- Use of Artificial Intelligence (AI) can help in disease prediction, diagnosis support, and decision-making.

- The system can be scaled to support multiple hospitals, making it suitable for large healthcare organizatio

ABOUT IT COMPANY

1. Tata Consultancy Services (TCS)

a) Company Overview

- Founded: 1968
- Headquarters: Mumbai, Maharashtra, India
- Website: www.tcs.com
- About: Tata Consultancy Services (TCS) is one of the largest global IT services, consulting, and business solutions organizations. It provides services in software development, cloud computing, artificial intelligence, and data analytics to businesses and educational institutions worldwide.

b) Education & Digital Services

TCS develops various education technology and digital management solutions:

- University ERP systems for managing student data and academic records
- Digital learning platforms and online education solutions
- Data analytics systems for student performance tracking
- Cloud-based education management systems
- Digital transformation solutions for universities and institutions

c) Technologies Used

- Artificial Intelligence & Machine Learning
- Data Analytics & Big Data
- Cloud Computing (AWS, Azure)

- Web Application Development
- Database Management Systems (MySQL, Oracle)

d) Career & Work Culture

- Focus on innovation, digital transformation, and global IT services
- Provides opportunities to work on enterprise-level software solutions
- Encourages continuous learning and professional development
- Offers training programs for fresh graduates and IT professionals

e) Placement Process

- Eligibility: BCA, B.Tech, BSc IT, MCA, and related IT streams with minimum 60%
- Rounds:
    1. Online aptitude test (logical reasoning, quantitative aptitude, English)
    2. Technical interview (programming, DBMS, projects)
    3. HR interview (communication skills, teamwork, personality)
- Focus Areas: Programming basics, problem-solving skills, and project knowledge.

f) Contact Information

- Website: https://www.tcs.com
- Email: corporate.communications@tcs.com

2. Infosys

a) Company Overview

- Founded: 1981
- Headquarters: Bengaluru, Karnataka, India
- Website: www.infosys.com
- About: Infosys is a global leader in technology services and consulting. The company helps organizations transform their businesses through digital platforms, AI, data analytics, and cloud computing solutions.

b) Education & Digital Services

Infosys develops many digital platforms and ERP solutions used in the education and enterprise sectors:

- Education management systems for universities
- Digital platforms for student data management
- AI-based analytics for performance tracking
- Cloud-based ERP solutions for organizations
- Learning management systems and training platforms

c) Technologies Used

- Artificial Intelligence & Machine Learning
- Data Analytics and Big Data Technologies
- Cloud Computing (Azure, AWS)
- Web Development Technologies
- Database Management Systems

d) Career & Work Culture

- Focus on innovation, digital transformation, and skill development
- Provides training through Infosys Global Education Center in Mysuru
- Encourages research, learning, and collaboration
- Offers opportunities to work on global technology projects

e) Placement Process

- Eligibility: BCA, BSc IT, B.Tech, MCA, and related IT streams with minimum academic performance requirements
- Rounds:
    1. Online aptitude test (logical reasoning, math, English)
    2. Technical interview (programming languages, DBMS, project discussion)
    3. HR interview (communication, teamwork, career goals)
- Focus Areas: Coding skills, analytical thinking, and project understanding.

f) Contact Information

- Website: https://www.infosys.com
- Email: info@infosys.com

CODE

```
#include <iostream>
#include <vector>
#include <fstream>
using namespace std;

class Patient {
public:
    int id, age;
    string name, disease, doctor, medicine;

    void addPatient() {
        cout << "Enter Patient ID: ";
        cin >> id;
        cin.ignore();

        cout << "Enter Name: ";
        getline(cin, name);

        cout << "Enter Age: ";
        cin >> age;
        cin.ignore();

        cout << "Enter Disease: ";
        getline(cin, disease);

        cout << "Enter Doctor Name: ";
        getline(cin, doctor);

        cout << "Enter Medicine: ";
        getline(cin, medicine);
    }
```

```
    void displayPatient() {
        cout << "\nID: " << id
             << "\nName: " << name
             << "\nAge: " << age
             << "\nDisease: " << disease
             << "\nDoctor: " << doctor
             << "\nMedicine: " << medicine << endl;
    }

    void writeToFile(ofstream &out) {
        out << id << endl
            << name << endl
            << age << endl
            << disease << endl
            << doctor << endl
            << medicine << endl;
    }

    void readFromFile(ifstream &in) {
        in >> id;
        in.ignore();


        getline(in, name);



        in >> age;
        in.ignore();


        getline(in, disease);
        getline(in, doctor);
        getline(in, medicine);
    }
};
```

```
int main() {
    vector<Patient> patients;
    int choice;

    // Load data from file
    ifstream infile("patients.txt");


    while (infile) {

        Patient p;
        p.readFromFile(infile);

        if (infile)
            patients.push_back(p);
    }

    infile.close();

    do {
        cout << "\n----- MEDICAL INFORMATION SYSTEM -----\n";
        cout << "1. Add Patient\n";
        cout << "2. Display All Patients\n";
        cout << "3. Search Patient\n";
        cout << "4. Delete Patient\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1) {
            Patient p;
            p.addPatient();
            patients.push_back(p);

            cout << "Patient added successfully!\n";
```

```
}

else if (choice == 2) {
    if (patients.empty()) {
        cout << "No records found!\n";
    } else {
        for (auto &p : patients) {
            p.displayPatient();
        }
    }
}



else if (choice == 3) {
    int searchId;
    cout << "Enter Patient ID to search: ";
    cin >> searchId;

    bool found = false;

    for (auto &p : patients) {
        if (p.id == searchId) {
            p.displayPatient();
            found = true;
            break;
        }
    }

    if (!found)
        cout << "Patient not found!\n";
}

else if (choice == 4) {
    int deleteId;
```

```
            cout << "Enter Patient ID to delete: ";
            cin >> deleteId;

            bool found = false;

            for (int i = 0; i < patients.size(); i++) {
                if (patients[i].id == deleteId) {
                    patients.erase(patients.begin() + i);
                    cout << "Patient deleted successfully!\n";
                    found = true;
                    break;
                }
            }

            if (!found)
                cout << "Patient not found!\n";
        }

    } while (choice != 5);

    // Save data to file
    ofstream outfile("patients.txt");

    for (auto &p : patients) {
        p.writeToFile(outfile);
    }

    outfile.close();

    cout << "Data saved to file. Exiting...\n";
    return 0;

}
```

OUTPUT

```
----- MEDICAL INFORMATION SYSTEM -----
1. Add Patient
2. Display All Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter choice:
```

```
Enter choice: 1
Enter Patient ID: 707
Enter Name: naina
Enter Age: 18
Enter Disease: diabetes
Enter Doctor Name: dr rahul gupta
Enter Medicine: melformin
Patient added successfully!
```

```
Enter choice: 3
Enter Patient ID to search: 707

ID: 707
Name: naina
Age: 18
Disease: diabetes
Doctor: dr rahul gupta
Medicine: melformin
```

```
----- MEDICAL INFORMATION SYSTEM -----
1. Add Patient
2. Display All Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter choice: 4
Enter Patient ID to delete: 707
Patient deleted successfully!
```

```
----- MEDICAL INFORMATION SYSTEM -----
1. Add Patient
2. Display All Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter choice: 5
Data saved to file. Exiting...


...Program finished with exit code 0
Press ENTER to exit console.
```

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